



Essay / Assignment Title: Computer Science for Digital Engineering.

Programme title: BSc Computer Science and Digitisation

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Year: 2026

CONTENTS

INTRODUCTION	4
CHAPTER ONE: SAP IN THE DIGITAL ECONOMY	5
1.1 GAME CONCEPT AND DESIGN LOGIC	5
1.2 BUILDING THE ENVIRONMENT AND THE THREE-DIMENSIONAL EFFECT.....	5
1.3 PUZZLE DESIGN AND THE REVEAL MECHANISM	7
1.4 QUESTION LOGIC AND ANSWER HANDLING	8
1.5 SCENE PROGRESSION AND BONUS FEATURES.....	9
CHAPTER TWO: MOONLIGHT REGALIA (FREECAD)	11
2.1 THEME AND INSPIRATION	11
2.2 METHOD: FROM 2D SKETCHES TO 3D BODIES	11
2.3 THE FOUR PIECES	13
2.4 THE PRESENTATION SCENE	15
2.5 THE COLLECTION IN A FANTASY SETTING	16
CONCLUDING REMARKS	17
BIBLIOGRAPHY	18
APPENDIX A: GITHUB REPOSITORY LINK	19

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Date: 25/09/2026

INTRODUCTION

The present report describes the design and the development of the two projects required under the practical brief – the interactive puzzle quiz game coded in C++ with OpenGL and the fantasy jewellery collection designed in FreeCAD. It is worth noting that I purposefully created the two works based on one visual language, the namely the polar sky, so that the submission will represent one cohesive body of work. Indeed, in the first project, users have to restore the photographic images of the aurora in the sky above a distant Arctic observatory. At the same time, the second project presents the transformation of the same visual theme into the jewellery collection. It means that the goals of each of the project are identical, as the task was to create convincing scenery with the help of the simple and recognizable geometric primitives, to maintain a clean construction history and to introduce sufficient interactive details.

CHAPTER ONE (Polar Skies (C++ and OpenGL))

1.1 Game concept and design logic

In Polar Skies, the player assumes the role of a night shift researcher at an Arctic observatory, whose telescope archives were broken down by a storm. In the process of fixing the archive plates, the player solves the picture puzzles where answering a question on the polar science topic correctly will restore the one puzzle piece out of the six. I chose such a concept because of the two reasons. Firstly, the northern lights is the only amazing phenomenon which can be portrayed using the line-art style required in the prompt. Secondly, the scenario allows to integrate the both quizzes into it, the first six questions about the northern lights and the night sky, the second six about the geography and the animals in the Arctic and all answers are accompanied by the additional facts so that the player always learns something. The correctnesses of the questions was verified based on the public sources on science (NOAA Space Weather Prediction Center, 2025).

1.2 Building the environment and the three-dimensional effect

The game takes place on a static logical screen consisting of the 1280 by 720 units projected in the orthographic projection into the window, hence coordinates in the code are independent of the resolution. Scene is rendered back-to-front in the layers as it was prescribed in the classic 2D pipeline of the Hearn, Baker and Carithers (2011). The sky is a single quad with the vertex colours that are interpolated by the OpenGL into a smooth night gradient. On top of this I place a roughly one hundred and thirty shining stars whose luminosity varies in the accordance with the offset sine curves, and the three aurora curtains made of the translucent quad strips that follow two superimposed sine waves; all of those curtains are brightly lit at their lower edge and the fade to black at their upper edge – it creates convincing effect of curtain light. Moon is constructed in the

layers from circles: a halo of the dimmed circles, face of the moon, darker circles for craters, a shaded terminator and an illuminating edge, and therefore a flat circle becomes spherical. This technique of constructing the three dimensional illusion through the layers works elsewhere in the scene as well: the observatory wall is rendered as eight vertical stripes, their shades being defined by a cosine function to make the flat wall appear cylindrical, the dome is shaded by lunes, all the objects have soft drop shadows consisting of the translucent ellipses, and there are two different toned mountain ridges creating parallax. The game is animated with one hundred and fifty snowflakes floating down, the flickering window light, blinking radio beacon, and slowly waving aurora. The scene is named-engraving according to the requirement by having a wooden post with SIDDHANT DANE engraving standing next to the station, and by HUD crediting the observer by name (Figure 1).



Figure 1: Level 1, the Polar Night. Aurora curtains, layered moon, shaded observatory, engraved signpost and the frosted archive plate.

1.3 Puzzle design and the reveal mechanism

Each archive plate is a small drawing made from the primitives; Plate I consists of the Great Bear over an ice fjord, the seven main stars of the Big Dipper connected by constellations lines above reflections of auroras in the still water, while the Plate II depicts a mother polar bear and her cub on an ice floe below the midnight sun. The plate is housed in an ice frame with a three by two matrix. The unveiling technique utilizes the OpenGL scissor test: for each tile discovered, the entire plate is rendered with clipping to the rectangle of the tile, thus ensuring the perfect fitting, just like a puzzle putting itself together. Unfound tiles are painted as frosty ice with rounded edges and stamped with a snowflake mark, while a new piece appears as if it is developing out of white like a photo developing in the fixer (Figure 2).



Figure 2: three of six pieces recovered; the scissor-clipped reveal keeps every piece perfectly registered.

1.4 Question logic and answer handling

The questions are the bare C++ structs that include the text, three options, the correct index and the field note, categorised by the levels into banks and then dequeued (Stroustrup, 2013). The game is a simple finite state machine with the states for the menu, the play, the feedback, completing the level, completing the game and the game over state, that makes the input handling and rendering honest: every state renders only its contents and the processes only its keys. A correct answer restores a piece and awards the player one hundred points plus twenty-five for a streak bonus up to the one hundred. An incorrect answer or running out of the twenty second timer costs the player a heat cell and twenty-five points, highlights the borders of the screen red, shows the correct answer with the field note and silently moves the failed question to the bottom of the queue, making it possible to complete the plate despite mistakes. Running out of all the three heat cells triggers the whiteout and causes the player to retry the level (Figure 3). The timer is rendered as a colour interpolating bar from green to red according to the amount of time remaining.

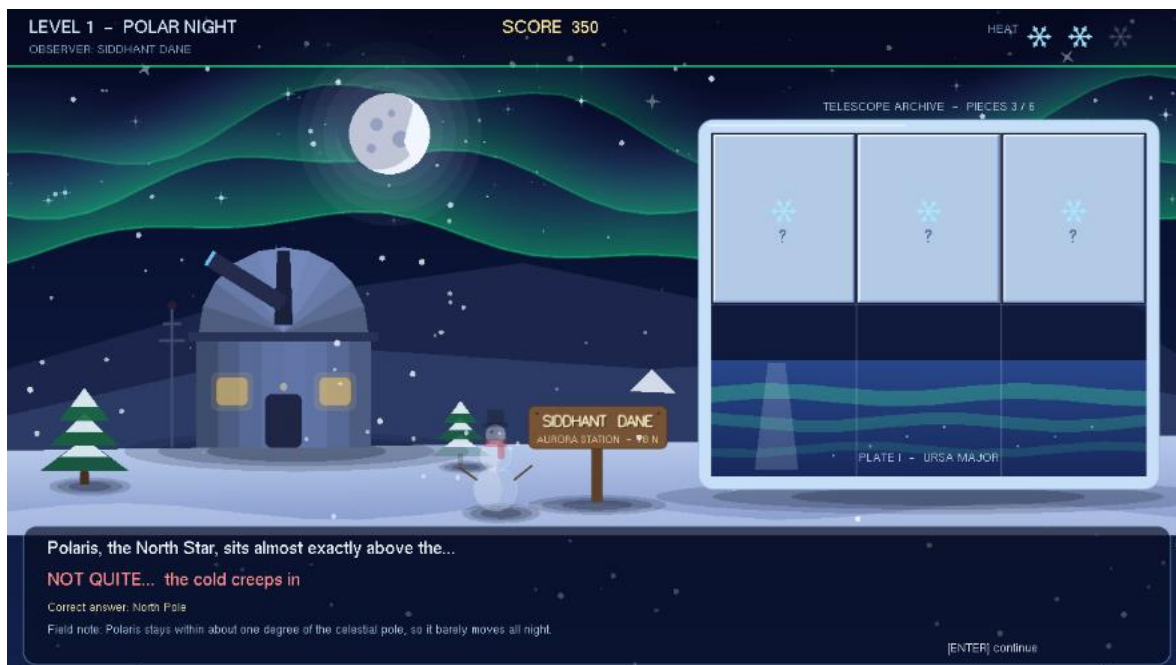


Figure 3: incorrect-answer handling; the correct answer and a field note are shown and the question returns later.

1.5 Scene progression and bonus features

Restoring the Plate I brings the end of the polar night: at the second level, all the colours switch to rose and violet polar dawn, the moon becomes a low sun with light tails, aurora calms down and there appears a completely new plate, meeting the requirement of progress with both new background and new puzzle (Figure 4). The restoration of Plate II results in a completion window with the star burst celebrations, a final score and a rank you got: from the Trainee Observer to Chief Aurora Scientist (Figure 5). In addition to the minimum requirements, the build offers continuous animation, the timer, the scoring system, heat cells with lives, colour feedback flashes, a key to freeze the timer, as well as a key to write a screenshot to the disk, that is how the images in this section were obtained. The project consists of one file compilable on the Windows, the Linux and macOS without any modifications using freeglut or GLUT library (Kessenich, Sellers and Shreiner, 2016; The freeglut Project, 2024).

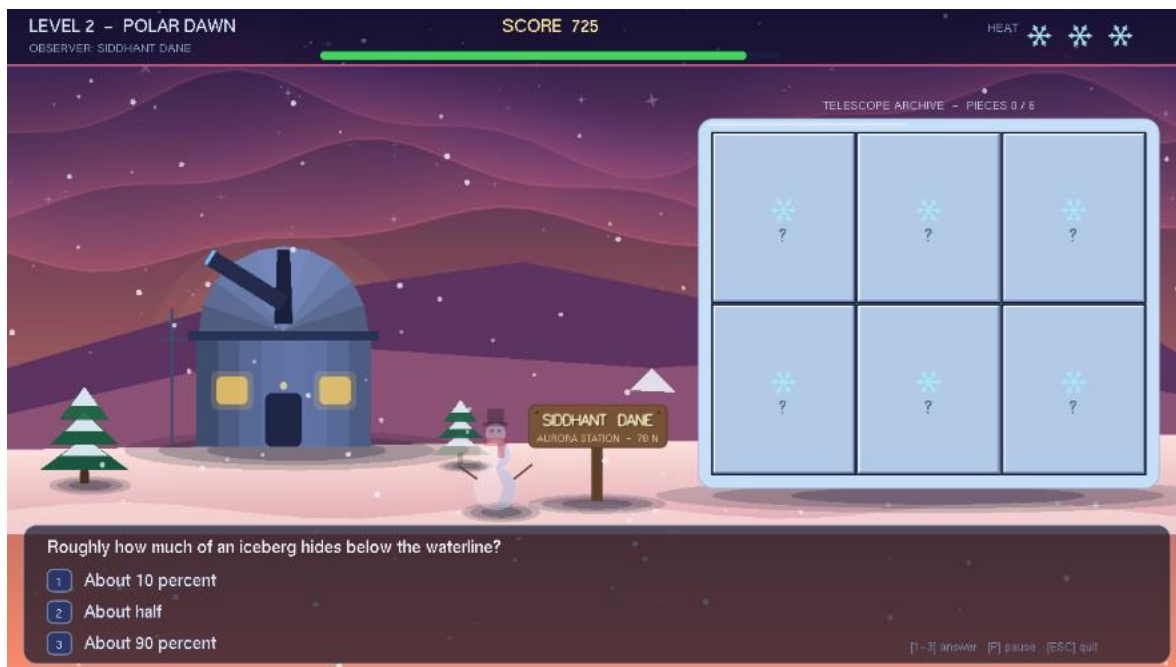


Figure 4: Level 2, the Polar Dawn - a new background scene and a new plate after the first is restored

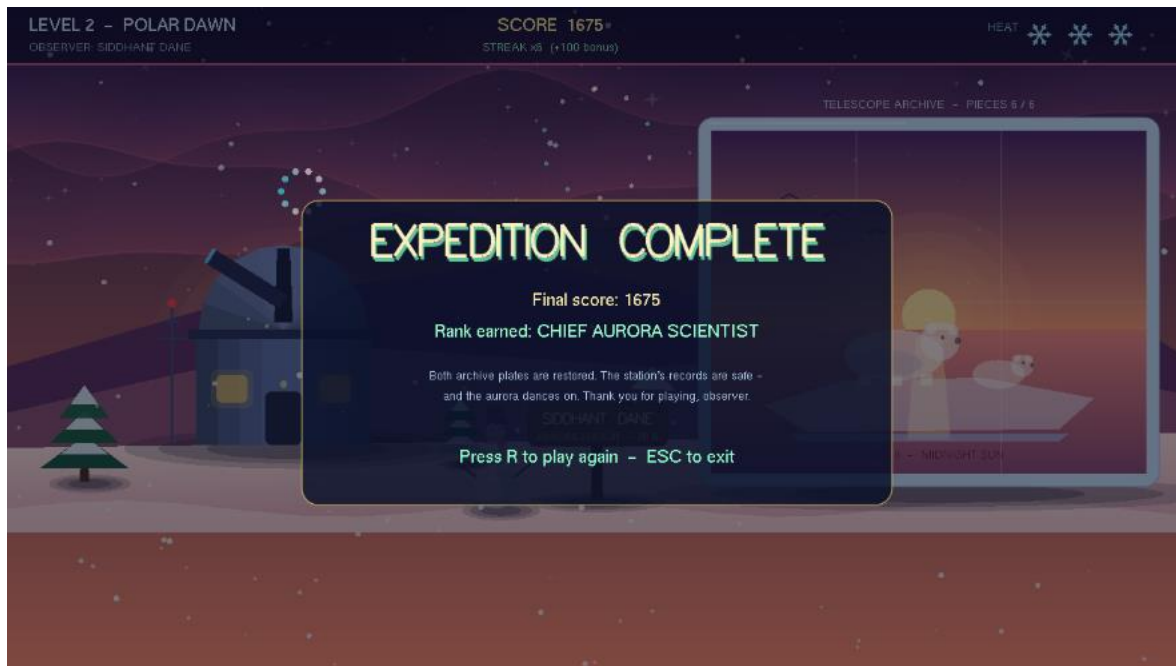


Figure 5: the completion screen with final score and rank.

CHAPTER TWO (Moonlight Regalia (FreeCAD))

2.1 Theme and inspiration

Moonlight Regalia is the royal jewellery of the fictitious court of the moon elves, and it consciously wears the same sky as in the Task 1. It consists of the four pieces: Selene's Band – a golden ring with a crescent-engraved bezel, the Star-forged Circlet – a silver crown of eight star-crested spikes, the Aurora Tear – a faceted tear of aurora-teal hanging from the crescent, and the Starlight Studs – a pair of earrings adorned with star and the pearl motifs. The vocabulary of the design is quite limited, which makes the collection a collection: all items in the set are composed exclusively of the crescents, five-pointed stars, teardrops and the spheres in silver, moon-gold, pearl, and one aurora-teal stone.

2.2 Method: from 2D sketches to 3D bodies

Everything starts out as the two-dimensional sketch geometry which is turned into a solid as per the brief, using the Sketcher and the Part Design modules (The FreeCAD Team, 2025). Revolution, Pad and Pocket and the Polar Pattern are the key operations, with everything revolving or patted out from a plane. There are two recurring constructions. First, a reusable crescent, constructed by using two circles that intersect each other by the retaining the outer arc of one and the facing arc of the other, showing up as a cut-out on the ring bezel at the two millimetres, as a pad on the pendant cradle at the eight and as a seventy-eight-millimetre pad in the background of the presentation scene – a wonderful example of the parametric thinking. Second, the revolved profile. All ring band, circlet band and pendant gem profiles are sketched as a single line which is then revolved around an axis – the gem being sketched as a six-point polyline which, when revolved, results in the stepped silhouette of a briolette-cut stone.

Everything is modelled to scale, with a seventeen-millimetre ring bore against a one-hundred-and-thirty-six-millimetre circlet ring, resulting in proper proportions in the presentation scene. Figure 6 is a crescent sketch open in the Sketcher, and

the Figure 7 is the completed model tree, which describes the process through named features.

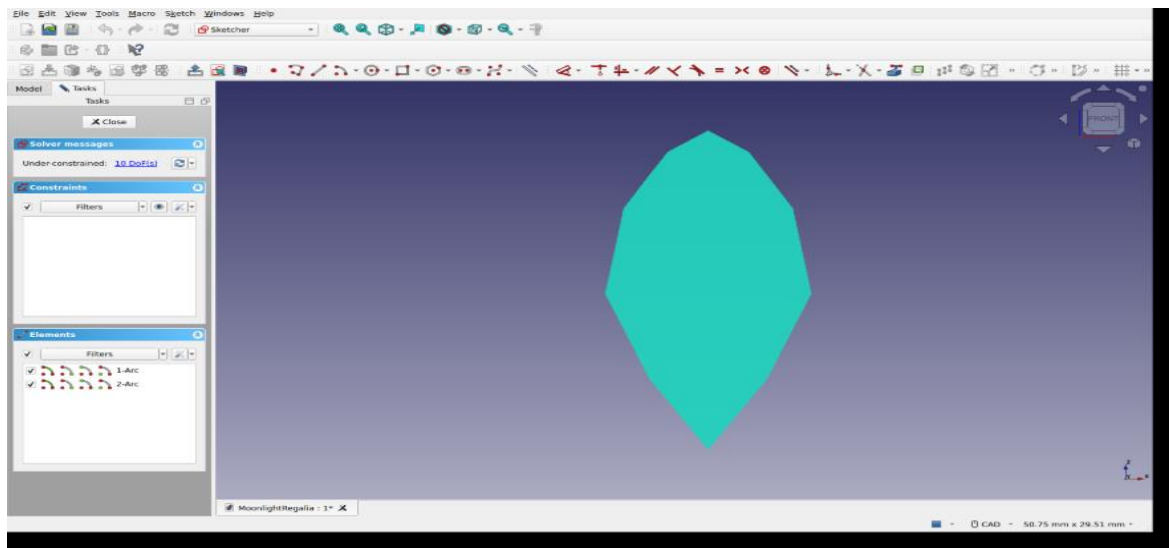


Figure 6: the pendant cradle's two-arc crescent sketch open in the FreeCAD Sketcher.

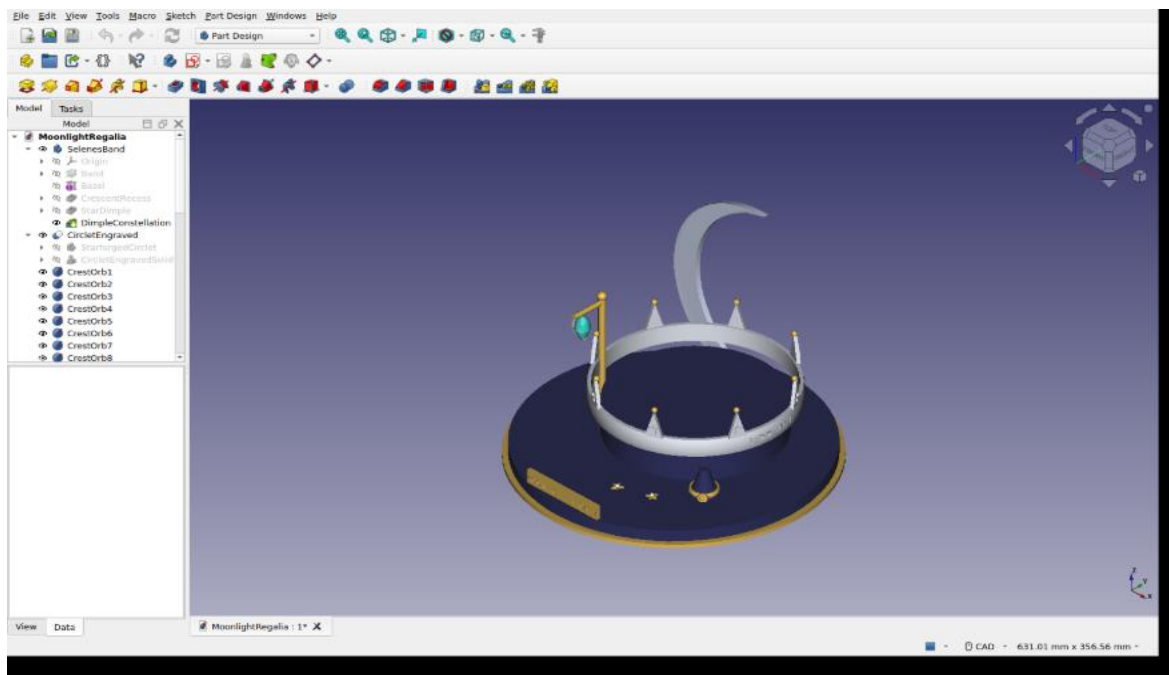


Figure 7: the finished document in FreeCAD 1.0 with the named feature history expanded

2.3 The four pieces

Selene's Band uses comfort-fit profile, flat inner, two shoulders and one bulged outer arc as inputs to revolve the band, followed by the creating round bezel from the shoulder. Crescent moon is inset within the bezel face, and a single star dimple pocketed eight times in a polar pattern gives a constellation which shadows and not the stones (Figure 8). The Star-forged Circlet uses a taller band and a polar pattern to sketch the two figures eight times each - a spike and a star crest, followed by the eight moon-gold orbs positioned on the spikes. Front of the band is devoid of the spikes as it contains the personalized text which is accomplished by using a ShapeString and carving 'SIDDHANT' in the curved band like an engraving done by a jeweller (Figure 9). The Aurora Tear requires two bodies - teal briolette and a silver frame, created by revolving the padded crescent cradle standing proud of the stone, a torus bail and three granulation beads positioned on the arc. The Starlight Studs require only a padded star plate, a pearl and a post; and the second earring is just an App Link of the first (Figure 10).



Figure 8: Selene's Band - revolved band, crescent-pocketed bezel and the polar-patterned star dimples.

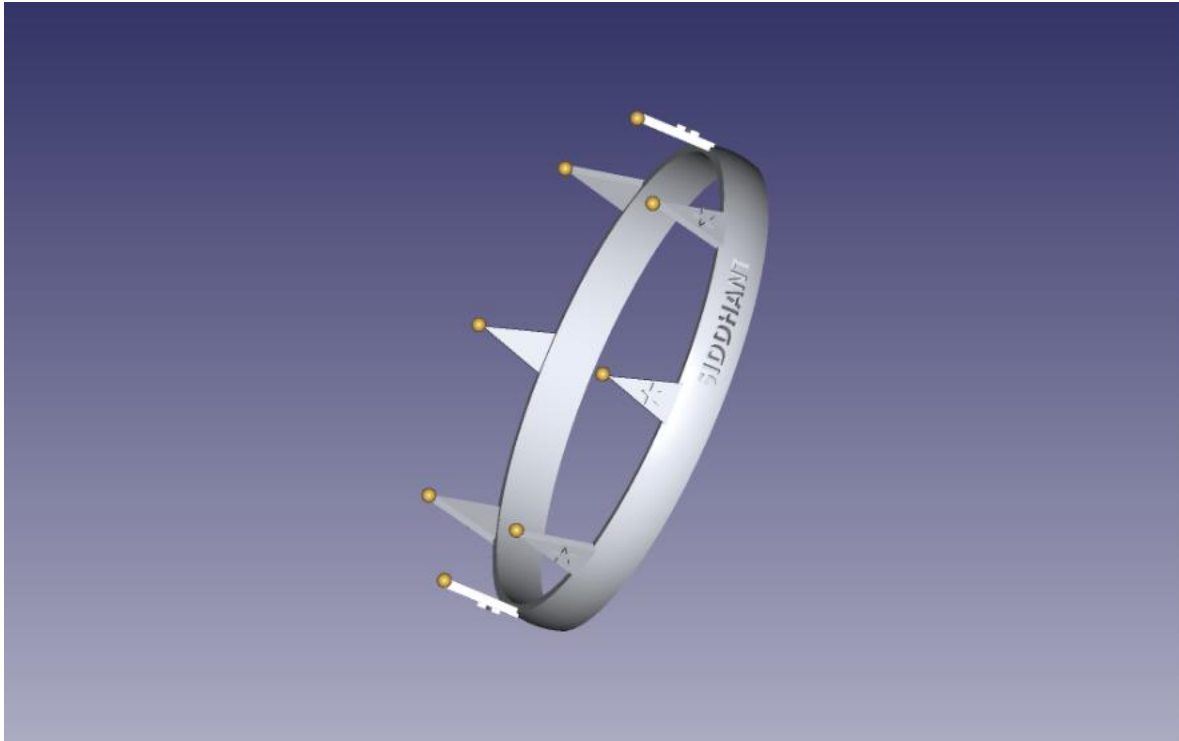


Figure 9: the Star-forged Circlet with the engraved name on the spike-free front of the band.

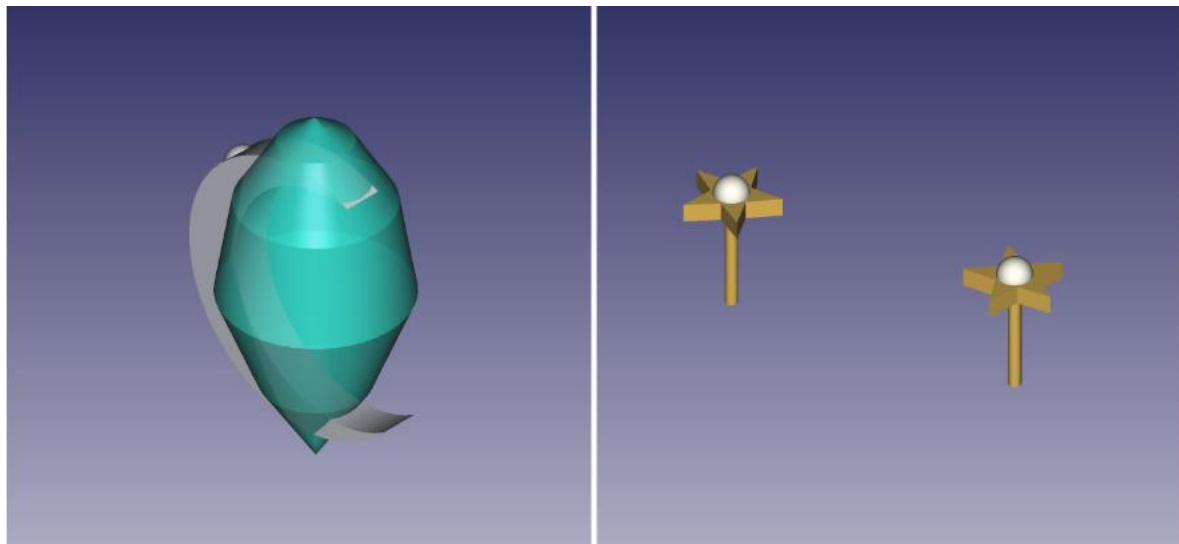


Figure 10: the Aurora Tear pendant and the Starlight Studs.

2.4 The presentation scene

All the four artifacts have been placed upon a jeweller's platform designed specifically for this purpose: the gold base topped with blue velvet, drum riser which supports the circlet, a cone for the ring, a hanging stand which has a rod which penetrates the bail of the pendant, and the nails which hold down the studs on the velvet. Behind the crown, the crescent shaped backing stands with its bottom end inserted into the base, and an engraving on the upper edge gives the name of the collection. Since all objects have their own colors, and since the geometry of the construction is not visible, the file opens with the finished picture (Fig. 11).



Figure 11: the final rendered presentation scene from the submitted FreeCAD file.

2.5 The collection in a fantasy setting

Regalia's fictional universe sees the Regalia as the coronation jewelry: the circlet crowns the Moonlit Sovereign, the Aurora Tear belongs to the astronomer in court as a container of the collected auroras, the band seals lunar edicts, and the studs belong to the new star-wardens. In practical terms, this same file can be used as a portfolio for the game design asset illustration as well as the beginning of miniatures design because all the solids are watertight and modelled in actual scale.

CONCLUDING REMARKS

Both the deliverables address the brief, go beyond the brief but stay within its parameters; an interactive quiz game, entirely made up of the pixels drawn using primitive geometries, and a four-piece parametric jewellery set with a clean sketch-first background, personalization and a final staged render. The work has helped me learn my boundaries. In the OpenGL, the hard part isn't math but rather discipline; a state machine and a predefined logical space kept the interface, the timer and the animations separate from each other. In FreeCAD, the lessons lay in parametric hygiene, knowing which arc to use in case of two intersecting circles, having pattern features on the tips of the bodies and orienting the text so that it looks right on a curved surface. Verification took place in the both cases, either by programmatically going through each and every state of the game or by validating the solid and the volume of each body, respectively.

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APPENDIX (Github Repository link)



Github Repository link: <https://github.com/SiddhantDane/Computer-Science-for-Digital-Engineering-Polar-Skies-Moonlight-Regalia>